# MagicBox Installation Guide

Goal: Portainer reachable from the LAN at **https://magicbox.local:9443/**, with the `homelab` environment and the default stacks (RabbitMQ, PostgreSQL, Ollama, Grafana + Loki + Alloy). MagicBox has no fixed IP: it is always reached as `magicbox.local` (mDNS via Avahi). `magicbox$` = on the Mini PC, `controller$` = on your workstation in this repository. Replace `<...>` with your values.

## 1. Manual Steps (Mini PC)

### 1.1 Before you start

- [ ] Back up anything on the internal disk (the OS install erases it).
- [ ] Have: USB stick (4 GB+), Ethernet cable to the LAN (recommended) or the Wi-Fi name and password, router admin access.
- [ ] Controller and Mini PC on the same subnet (not guest Wi-Fi).

### 1.2 Firmware (UEFI)

Setup key is usually `Del`, boot menu `F7` (check the boot screen).

- [ ] Apply a GMKtec firmware update for your model, if one exists.
- [ ] Boot mode: UEFI. Secure Boot: enabled.
- [ ] *Restore on AC Power Loss*: **Power On**.
- [ ] Save and exit.

### 1.3 Install Ubuntu Server 24.04 LTS

(Also supported: Ubuntu 26.04, Debian 12/13.)

- [ ] Download the ISO from `https://ubuntu.com/download/server`, verify its SHA-256, write it to the USB stick, boot from it.
- [ ] Network: wired or Wi-Fi, DHCP (no static IP). Skipped Wi-Fi: see 1.3.1.
- [ ] Storage: entire internal disk, **no** passphrase encryption (it blocks unattended reboots).
- [ ] Server name: **`magicbox`**. Create your admin user (`<admin-user>`).
- [ ] Tick **Install OpenSSH server**. Featured snaps: **none**.
- [ ] Reboot, remove the USB stick, log in.

#### 1.3.1 Wi-Fi (only if skipped during install)

Replace the three `<...>` values (`ip -br link` shows the Wi-Fi interface, usually starting with `wl`) and run:

```
magicbox$ sudo netplan set --origin-hint 60-wifi 'wifis.<wifi-interface>={dhcp4: true,
access-points: {"<wifi-name>": {password: "<wifi-password>"}}}'
magicbox$ sudo netplan apply
```

The first command writes `/etc/netplan/60-wifi.yaml` (readable by root only). If the name or password contains `'`, `"` or `\`, edit that file instead.

### 1.4 Update and collect facts

```
magicbox$ sudo apt update && sudo apt full-upgrade -y && sudo reboot
magicbox$ hostname                          # must print magicbox
magicbox$ cat /etc/os-release; uname -m     # supported OS, x86_64
magicbox$ ip -br link                       # LAN interface name
magicbox$ ip -br -4 addr                    # LAN subnet (e.g. 192.168.1.x/24)
```

### 1.5 Avahi (magicbox.local)

```
magicbox$ sudo apt install -y avahi-daemon avahi-utils libnss-mdns
magicbox$ sudo systemctl enable --now avahi-daemon
magicbox$ avahi-resolve-host-name -4 magicbox.local
```

- [ ] The last command prints `magicbox.local` and an IPv4 address.
- [ ] From the controller: `ping -c 3 magicbox.local` answers. (Linux controller: `sudo apt install avahi-daemon libnss-mdns` first.)

### 1.6 SSH key

```
controller$ mkdir -m 700 -p ~/.ssh/magicbox
controller$ ssh-keygen -t ed25519 -f ~/.ssh/magicbox/id_ed25519 -C magicbox
controller$ ssh-copy-id -i ~/.ssh/magicbox/id_ed25519.pub <admin-user>@magicbox.local
controller$ ssh -i ~/.ssh/magicbox/id_ed25519 <admin-user>@magicbox.local 'sudo -v && echo
sudo-ok'
```

- [ ] Dedicated key for MagicBox: private `~/.ssh/magicbox/id_ed25519` (never share it), public `~/.ssh/magicbox/id_ed25519.pub`.
- [ ] Last command prints `sudo-ok`.
- [ ] Optional, after key login works (keep a session open while testing):

```
magicbox$ echo 'PasswordAuthentication no' | sudo tee /etc/ssh/sshd_config.d/10-magicbox.conf
magicbox$ sudo sshd -t && sudo systemctl reload ssh
```

### 1.7 Values for the inventory

| Value | Source | Yours |
|---|---|---|
| ansible_host | always magicbox.local | magicbox.local |
| ansible_user | admin user | _____ |
| ansible_ssh_private_key_file | key from 1.6 | ~/.ssh/magicbox/id_ed25519 |
| ansible_become_exe | Ubuntu 26.04 only | sudo.ws |
| magicbox_lan_interface | ip -br link | _____ |
| magicbox_lan_cidr | LAN subnet, e.g. 192.168.1.0/24 | _____ |

## 2. Controller setup

- [ ] Install `uv` (`brew install uv`), `git`, `make`. Windows: use WSL2.

```
controller$ make setup
controller$ make check
```

## 3. Configuration

```
controller$ cp inventory/hosts.example.yml inventory/hosts.yml
controller$ uv run ansible-vault create inventory/group_vars/magicbox/vault.yml
```

- [ ] In `inventory/hosts.yml`: set `ansible_user`, `magicbox_lan_interface`, `magicbox_lan_cidr` (1.7). Keep `ansible_host: magicbox.local` and `ansible_ssh_private_key_file: ~/.ssh/magicbox/id_ed25519`.
- [ ] Ubuntu 26.04: uncomment `ansible_become_exe: sudo.ws`.
- [ ] In `vault.yml` (see `inventory/vault.example.yml`): `vault_become_password` (sudo password of `ansible_user`) and four different passwords of 12+ characters: `vault_portainer_admin_password`, `vault_postgresql_password`, `vault_rabbitmq_password`, `vault_grafana_admin_password`. Keep the vault password in your password manager.

```
controller$ make secrets-check
controller$ uv run ansible magicbox -m ansible.builtin.ping    # expect "pong"
```

## 4. Deployment

```
controller$ make deploy EXTRA="-e magicbox_confirm_destructive=true"   # first run
controller$ make deploy                                                # later runs
```

`make` asks only for the vault password; the sudo password comes from `vault_become_password`. The first run needs the confirmation flag because it enables the firewall. The CA certificate is saved to `artifacts/magicbox-local-ca.crt`. Portainer is initialised with the `homelab` environment.

Then deploy the default stacks (the first run pulls about 4 GB of images):

```
controller$ make stacks
```

Deploy only some stacks:

```
controller$ uv run ansible-playbook playbooks/stacks.yml --ask-vault-pass -e
'{"stacks_enabled": ["postgresql"]}'
```

## 5. Verification

```
controller$ make verify
controller$ make idempotency VAULT_PASSWORD_FILE=<file>
```

- [ ] `make verify` passes.
- [ ] `make idempotency` ends with "Idempotency check passed". Keep the password files outside the repo with mode 0600; delete them afterwards.
- [ ] From another LAN computer: `ping magicbox.local` works.
- [ ] Browser (CA trusted, section 6): `https://magicbox.local:9443/` opens with no warning; log in as `admin` with the vault password.
- [ ] `sudo reboot` on MagicBox: the page comes back, settings kept.
- [ ] `magicbox$ sudo ss -ltn | grep -E ':(9000|2375|2376) '` prints nothing.
- [ ] Portainer > Environments shows `homelab`; Stacks shows `rabbitmq`, `postgresql`, `ollama`, `observability`.
- [ ] A second `make stacks` reports `changed=0`.

Services (from the LAN):

| Service | Address | Login |
|---|---|---|
| Grafana | https://magicbox.local:3000/ | admin / vault_grafana_admin_password |
| RabbitMQ UI | http://magicbox.local:15672/ | magicbox / vault_rabbitmq_password |
| RabbitMQ AMQP | amqp://magicbox.local:5672 | same as UI |
| PostgreSQL | magicbox.local:5432, database magicbox | magicbox / vault_postgresql_password |
| Ollama | http://magicbox.local:11434/ | none (LAN only) |

- [ ] Grafana > Explore > Loki: `{container="postgresql"}` shows logs.

## 6. TLS trust (each client, once)

- [ ] Check the fingerprints match:

```
controller$ openssl x509 -in artifacts/magicbox-local-ca.crt -noout -fingerprint -sha256
magicbox$ sudo openssl x509 -in /opt/magicbox/portainer/certs/ca.crt -noout -fingerprint
-sha256
```

- [ ] Copy `magicbox-local-ca.crt` to the client and install it:

| Client | Command / steps |
|---|---|
| macOS | sudo security add-trusted-cert -d -r trustRoot -k /Library/Keychains/System.keychain magicbox-local-ca.crt |
| Windows (admin PowerShell) | Import-Certificate -FilePath magicbox-local-ca.crt -CertStoreLocation Cert:\LocalMachine\Root |
| Ubuntu/Debian | sudo cp magicbox-local-ca.crt /usr/local/share/ca-certificates/ && sudo update-ca-certificates |
| Firefox | Settings > Certificates > View Certificates > Authorities > Import > trust for websites |
| iOS | Install the profile, then General > About > Certificate Trust Settings > enable |
| Android | Settings > Security > Encryption & credentials > Install a certificate > CA certificate |

- [ ] Restart the browser.

The CA only covers `magicbox.local` and `magicbox`. The server certificate is renewed automatically by any playbook run within 30 days of expiry.

### 6.1 If a client cannot resolve magicbox.local

- [ ] Client on the same LAN subnet (not guest Wi-Fi, not VPN, not another VLAN).
- [ ] macOS, iOS, Windows 10/11: built in. Android 12+: most browsers.
- [ ] Linux: `sudo apt install avahi-daemon libnss-mdns`.
- [ ] On MagicBox: `systemctl status avahi-daemon` is active.

## 7. Troubleshooting

| Symptom | Fix |
|---|---|
| "Timeout waiting privilege escalation prompt" | Ubuntu 26.04 (sudo-rs): set ansible_become_exe: sudo.ws |
| "Unsupported OS" | Install Ubuntu Server 24.04 LTS |
| magicbox_lan_interface assertion | Use a name listed in the error |
| Admin password assertion | Create vault.yml; run with --ask-vault-pass |
| "ufw is inactive" | Add -e magicbox_confirm_destructive=true |

| Symptom | Fix |
|---|---|
| Controller "outside" the LAN | Fix magicbox_lan_cidr or run from the LAN |
| Conflicting Docker packages | Check nothing uses them, then add the confirmation flag |
| Docker key fingerprint mismatch | Stop and verify Docker's key out of band |
| magicbox.local not resolving | systemctl status avahi-daemon; see 6.1 |
| No network over Wi-Fi | Check name/password in /etc/netplan/60-wifi.yaml; sudo netplan --debug apply |
| Name became magicbox-2.local | Rename the other magicbox device; restart avahi-daemon |
| Certificate warning | Install the CA (section 6); restart the browser |
| Connection refused | sudo docker ps -a; sudo docker logs portainer; sudo ufw status |
| Vault password rejected at login | Portainer was already initialised: reset (8.4) |
| stacks: "Portainer rejected the vault password" | Password changed in the UI: update vault_portainer_admin_password |
| stacks: environment not found | Run make deploy first |
| stacks: wait for containers times out | Slow image pull: re-run make stacks; check sudo docker ps -a |
| Stack service rejects the vault password | Password was set at first start; change it in the service, then in the vault |

# 8. Recovery

## 8.1 Repair drift

```
controller$ make deploy
```

## 8.2 Update packages

```
controller$ make update
controller$ make update EXTRA="-e os_update_reboot=true"   # if a reboot is reported
```

- [ ] Run monthly. The reboot stops Portainer and the stacks for a few minutes.

## 8.3 Firewall lockout

At the console: `sudo ufw disable`, fix `magicbox_lan_cidr` (IPv4 subnet), then redeploy with `-e magicbox_confirm_destructive=true`.

## 8.4 Lost admin password

```
magicbox$ sudo docker stop portainer
magicbox$ sudo docker run --rm -v portainer_data:/data portainer/helper-reset-password
magicbox$ sudo docker start portainer
```

## 8.5 Backup and restore

```
magicbox$ sudo docker stop portainer
magicbox$ sudo tar -C / -czf /root/magicbox-backup-$(date +%F).tgz opt/magicbox
var/lib/docker/volumes/portainer_data
magicbox$ sudo docker start portainer
```

Stack data lives in the volumes `postgresql_data`, `rabbitmq_data`, `grafana_data`, `loki_data` and `ollama_data` under `/var/lib/docker/volumes/`. Stop the stack in Portainer before copying a volume. For PostgreSQL, prefer a logical dump:

```
magicbox$ sudo docker exec postgresql pg_dumpall -U magicbox | gzip > magicbox-pg-$(date
+%F).sql.gz
```

The archive contains the CA key: store it encrypted, off the box. To restore, reinstall (sections 1-4), stop Portainer, run `sudo tar -C / -xzf <archive>`, and start Portainer.

## 8.6 Rotate the CA

Delete `/opt/magicbox/portainer/certs/` on MagicBox, run `make deploy`, and install the new CA on every client. Remove the old CA from the clients.